Binary Cross Entropy Loss function Differentiation

Binary cross entropy loss function

$$L = -[y\log(a) + (1-y)\log(1-a)]$$

$$\therefore dL/da = -\frac{d}{da}[y\log(a) + (1-y)\log(1-a)]$$

$$\frac{d}{da}[y\log(a)] = y/a \quad \text{---(i) [first term]}$$

$$\frac{d}{da}[(1-y)\log(1-a)] = (1-y)\cdot\frac{-1}{1-a} = -\frac{1-y}{1-a} \quad \text{---(ii) [Second term]}$$

Substituting in the main equation

$$dL/da = -\left(y/a - \frac{1-y}{1-a}\right) = -y/a + \frac{1-y}{1-a}$$

$$= \frac{-y+ay+a-ay}{a\cdot(1-a)}$$

$$= \frac{a-y}{a(1-a)}$$

Let

$$a = \sigma(z) = \frac{1}{1+e^{-z}}$$

$$\frac{da}{dz} = a(1-a)$$

Using Chain rule:

$$\frac{dL}{dz} = \frac{dL}{da} \cdot \frac{da}{dz}$$

$$= \frac{a-y}{a(1-a)} \cdot a(1-a)$$

$$= a-y$$

$$\boxed{\therefore \frac{dL}{dz} = a-y}$$

a = Predicted

y = actual